

## SUMMARY

I am an aspiring physician scientist excited to leverage the intersection of computer science, genomics, and neuro-immunology to better understand brain tumors and develop more effective therapies for patients.

## EDUCATION

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| 2026-2034 | Case Western Reserve University School of Medicine<br>Medical Scientist Training Program (MSTP) <ul style="list-style-type: none"><li>• MD (expected 2034)</li><li>• PhD in Systems Biology &amp; Bioinformatics (expected 2030)</li></ul> |
| 2019-2023 | University of Washington, Seattle <ul style="list-style-type: none"><li>• Bachelor of Science</li><li>• Major in Computer Science</li></ul>                                                                                                |

## HONORS AND AWARDS

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|------|------------------------------------------------------------------------------------|
| 2026 | 2026 Basic Science Award (Annual Meeting of the Society for Neuro-Oncology)        |
| 2026 | Best Poster Prize (Case Comprehensive Cancer Center Annual Retreat)                |
| 2025 | Best Basic Science Oral Presentation (CWRU Department of Pathology Annual Retreat) |
| 2022 | Washington Research Foundation Fellowship                                          |
| 2022 | Mary Gates Research Scholarship (University of Washington, Seattle)                |

## WORK EXPERIENCE

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| 2024-2026 | Computational Biologist, Case Western Reserve University, Cleveland, OH <ul style="list-style-type: none"><li>• <i>Investigated brain tumor genomics in the laboratory of Dr. Tyler E. Miller MD, PhD</i></li><li>• Built and maintained new cloud-based bioinformatics pipelines that analyze<ul style="list-style-type: none"><li>○ raw single-cell RNA-sequencing (scRNA-seq) data: performs quantification, QC, consensus non-negative matrix factorization (cNMF) to analyze cell states</li><li>○ raw spatial proteomics data: performs cell segmentation, marker positivity prediction, downstream correlative analyses</li></ul></li><li>• Analyzed multi-omics data (publicly available and novel) from brain tumor patients and murine models of rare cancers to characterize the immune microenvironment</li><li>• Mentored 1 undergraduate student, onboarded 5 computational lab members, and trained wetlab colleagues to interface with ‘omics data</li><li>• <i>Research project details are outlined in the following section.</i></li></ul> |
| 2023-2024 | Bioinformatics Researcher, Kallyope Inc, New York City, NY <ul style="list-style-type: none"><li>• <i>Kallyope Inc. is a drug discovery startup investigating diseases along the gut/brain axis</i></li><li>• Processed, clustered, and integrated single cell RNA-sequencing datasets from internal experiments and public databases to generate 40+ curated transcriptomic atlases</li><li>• Supported strategic decision-making across 4 major drug discovery programs by analyzing atlases and presenting findings to stakeholders<ul style="list-style-type: none"><li>○ biomarker discovery, organoid characterization, on-target off-site adverse effects</li></ul></li></ul>                                                                                                                                                                                                                                                                                                                                                                          |

- Spearheaded frontend & backend updates to internal software to (1) enable wetlab scientists to generate custom plots and (2) revamp system design for future features
- Fine-tuned a pre-trained vision transformer model to detect key points in behavioral assay videos of mice for novel behavior detection
- Built an end-to-end wrapper for a publicly available evolutionary algorithm that enabled medicinal chemists to generate new compounds using target receptor structures

## RESEARCH EXPERIENCE

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| 2024-Present | <p>Case Western Reserve University, Department of Pathology<br/>           PI: Tyler E. Miller MD, PhD<br/>           Project: “Single-cell Analysis of Adult and Pediatric CNS Tumors Reveals a Shared Framework of Immunomodulatory Myeloid Programs within the Tumor Microenvironment”</p> <ul style="list-style-type: none"> <li>● Assembled a scRNA-seq dataset of 1.5+ million cells from 213 patients by reprocessing raw, publicly available data of 8+ types of brain tumors (2-3 cohorts each)</li> <li>● Analyzed the immune populations across each tumor type, profiled their cell states using cNMF, and characterized a pan-tumor consensus set of gene expression programs</li> <li>● The cellular usage of these programs appear to be clinically and spatially remarkable</li> <li>● Led to numerous invited talks and poster presentations, won 2 conference awards</li> </ul> |
| Fall 2025    | <p>Case Western Reserve University, Department of Biomedical Engineering<br/>           PI: Chris Flask, PhD<br/>           Project: “Deep Learning Methods for Unbiased Segmentation of Tumor Architecture in Magnetic Resonance Fingerprinting (MRF) Studies of Murine Cancer Models”</p> <ul style="list-style-type: none"> <li>● Identified phenotypes in murine tumor MRF data by leveraging an autoencoder</li> <li>● Characterized the tumor brim and tumor core to improve upon a generalized hard threshold approach previously used for the data</li> <li>● Completed during a PhD rotation in Dr. Flask’s group</li> </ul>                                                                                                                                                                                                                                                             |
| 2025-2026    | <p>Case Western Reserve University, Department of Pathology<br/>           PI: Tyler E. Miller MD, PhD; Alex Y. Huang MD, PhD<br/>           Project: “TGF-<math>\beta</math> Regulates Oncogene Expression and the Immune Microenvironment in Osteosarcoma”</p> <ul style="list-style-type: none"> <li>● Analyzed scRNA-seq data for a 16 mouse cohort of osteosarcoma, studying effects of Vactosertib (TGF-<math>\beta</math> type 1 receptor kinase inhibitor) on tumor-laden &amp; healthy animals</li> <li>● Profiled the cell states of myeloid, T-cell, B-cell populations in the microenvironment and found meaningful correlations with tumor and treatment status that are currently being studied in the wetlab</li> <li>● Led to a grant proposal and ongoing manuscript</li> </ul>                                                                                                  |
| 2024-2025    | <p>Case Western Reserve University, Department of Pathology<br/>           PI: Tyler E. Miller MD, PhD; Alex Y. Huang MD, PhD<br/>           Project: “Piezo1 deletion enhances cross-priming of CD8<sup>+</sup> T cells by tumor-infiltrating CD11b<sup>+</sup> dendritic cells”</p> <ul style="list-style-type: none"> <li>● Analyzed scRNA-seq data for a 16 mouse cohort of rhabdomyosarcoma, studying the effects of conditional Piezo1 knockout in CD11b<sup>+</sup> and CXCL3<sup>+</sup> cells</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                 |

- Profiled cell states of myeloid, T-cell populations in the microenvironment and found significant enrichment in markers of antigen presentation in dendritic cells and activation in T cells from the knockout group
- Led to third-author publication in the Journal for Immunotherapy of Cancer

2020-2023

University of Washington, Department of Chemical Engineering

PI: David A.C. Beck, PhD; Mary E. Lidstrom, PhD

Project: “A Computational Atlas for Transcriptomic Exploration of *Methylovibrio buryatense* Using Unsupervised Machine Learning and Interactive Data Visualization”

- Integrated 10 years of bulk RNA-seq data studying the growth of methane-consuming bacterium *M. buryatense* to enable unsupervised clustering analysis
- Created a biologist-facing software tool to enable wetlab researchers to probe this dataset, its gene clusters, and enrichment analysis results
- Led to an undergraduate thesis, multiple oral and poster presentations, and won 2 prestigious student fellowships/awards.

Spring 2021

University of Washington, Department of Computer Science and Engineering

PI: Walter L. Ruzzo

Project: “Towards an Explainable Deep Learning-Based Exploration of Non-Coding RNA”

- Developed a feature attribution wrapper using Shapley Additive Explanations (SHAP) for a deep learning model that classifies function of non-coding RNA given sequence
- Built a data visualization pipeline to enable biologists to interpret feature attribution results overlaid on predicted ncRNA structure
- Collaborated with 3 students as my undergraduate Computational Biology Capstone

## INSTITUTIONAL SERVICE

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2026-Present

MSTP Retreat Committee, Case Western Reserve University, Cleveland, OH

- Elected by the student body of the CWRU MSTP program
- Collaborate with the team of 3 students, MSTP directors, and admin to plan & run a 2-day scientific conference (invite speakers, plan workshops, etc.)

2026-Present

APSA Midwest Regional Conference (Logistics Committee), Cleveland, OH

- Collaborate with the team of 7 students to plan & run a scientific conference for undergraduate students, medical students, residents, fellows and attending physicians interested in research.

2022-2023

Undergraduate Research Mentor, University of Washington, Seattle, WA

- Helped undergraduate students explore their research interests through conversation, narrow down labs, and strategize how to succeed as a researcher

## LEADERSHIP

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2020-2023

President, Impact++, University of Washington

- Organized a group of 80+ undergraduate students across 6 annual pro bono software projects for social good non-profits across the globe
- Trained developers, project managers, and designers to support projects
- Procured new projects with stakeholder negotiations across non-profit leaders, senior engineers/directors at tech orgs (Microsoft, startups, academic labs)

2021-2023

Journal Club Chair, Pre-Physician Scientist Club (PPSC), University of Washington

- Organized events where speakers across the training path (medical students, residents, fellows, and attending physicians) were invited to share their journey, research, and advice

## BIBLIOGRAPHY

### Peer Reviewed Articles

1. Bonner M, Askew D, **Sathish Kumar V**, Tomchuck SL, Eid S, Abiff M, Myers JT, Nguyen P, Garyu JWA, Miller TE, Huang AY. Piezo1 deletion enhances cross-priming of CD8<sup>+</sup> T cells by tumor-infiltrating CD11b<sup>+</sup> dendritic cells. *Journal for Immunotherapy of Cancer*. June 23, 2025; Volume 13: e011815. PMID: 40550569
2. Gonzalez Castro LN\*, **Sathish Kumar V\***, Suva ML, Miller TE. Redefining the immune microenvironment of gliomas in the era of single-cell genomics. *Neuro-Oncology Advances*. February 27, 2026; Volume 8: i4-i16. PMID: 41769417
3. Sloan AR, Bukenya G, Tannish GP, Silver DJ, ..., **Sathish Kumar V**, Jain S, Scalise A, ..., Miller TE, O'Connor CM, Nieman MT, Sangwan N, Chan TA, Khorana AA, Dhawan A, Cameron SJ, Lathia JD. Platelets drive immune suppression and glioblastoma growth in a sex-dependent manner via PAR4 signaling. *Cancer Cell*. August 4, 2026. (In Press)

### Oral Abstract Presentations

1. **Sathish Kumar V**, Wilson E, Lidstrom ME, Beck DAC. "A Computational Atlas for Transcriptomic Exploration of *Methylobacterium Buryatense* Using Unsupervised Machine Learning and Interactive Data Visualization"
  - a. International Conference on Computational Advances in Bio and Medical Sciences (ICCABS); Norman, OK (2023). (Oral presentation delivered by Sathish Kumar, V)
  - b. University of Washington Undergraduate Research Symposium; Seattle, WA (2023). (Oral presentation delivered by Sathish Kumar, V)
2. **Sathish Kumar V**, Estrada E, Revil T, Couturier C, Miller TE. "A Unified Framework for Immunosuppressive Myeloid Cells Across Brain Tumors"
  - a. Translational Research Cancer Centers Consortium (TRCCC); Seven Springs, PA (2025). (Oral presentation delivered by Sathish Kumar, V)
  - b. CWRU Department of Pathology Annual Retreat; Cleveland, OH (2025). (Oral presentation delivered by Sathish Kumar, V)
3. **Sathish Kumar V**, Estrada E, Revil T, Couturier C, Miller TE. "A Unified Framework for Immunomodulatory Myeloid Cells Across Brain Tumors"
  - a. Translational Research Cancer Centers Consortium (TRCCC); Seven Springs, PA (2026). (Oral presentation delivered by Sathish Kumar, V)
4. **Sathish Kumar V**, Estrada E, Revil T, Couturier C, Miller TE. "Single-cell Analysis of Adult and Pediatric CNS Tumors Reveals a Shared Framework of Immunomodulatory Myeloid Programs within the Tumor Microenvironment"
  - a. Annual Meeting of the Society of Neuro-Oncology; Philadelphia, PA (2026). (Oral presentation delivered by Sathish Kumar, V)
  - b. Cellular Plasticity & Tumor Ecosystem Conference; Cleveland, OH (2026). (Oral presentation delivered by Sathish Kumar, V)

### Poster Presentations

1. **Sathish Kumar V**, Wilson E, Lidstrom ME, Beck DAC. “A Computational Atlas for Transcriptomic Exploration of *Methylovibrio buryatense* Using Unsupervised Machine Learning and Interactive Data Visualization”
  - a. University of Washington Undergraduate Research Symposium; Seattle, WA (2023).
  - b. Paul G. Allen School of Computer Science Research Showcase; Seattle, WA (2023).
2. **Sathish Kumar V**, Estrada E, Revil T, Couturier C, Miller TE. “A Unified Framework for Immunosuppressive Myeloid Cells Across Brain Tumors”
  - a. Translational Research Cancer Centers Consortium (TRCCC); Seven Springs, PA (2025).
3. **Sathish Kumar V**, Estrada E, Revil T, Couturier C, Miller TE. “A Unified Framework for Immunomodulatory Myeloid Cells Across Brain Tumors”
  - a. Translational Research Cancer Centers Consortium (TRCCC); Seven Springs, PA (2026).
4. **Sathish Kumar V**, Estrada E, Revil T, Couturier C, Miller TE. “Single-cell Analysis of Adult and Pediatric CNS Tumors Reveals a Shared Framework of Immunomodulatory Myeloid Programs within the Tumor Microenvironment”
  - a. Case Comprehensive Cancer Center Annual Retreat (2026).

### **MEMBERSHIPS**

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Society of Neuro-Oncology

American Physician Scientist Association

**Languages:** fluent in English, Tamil

**Interests:** Tennis, Boulderling, Bass Guitar, Sci-Fi Movies, History of Science & Medicine, Board Games